

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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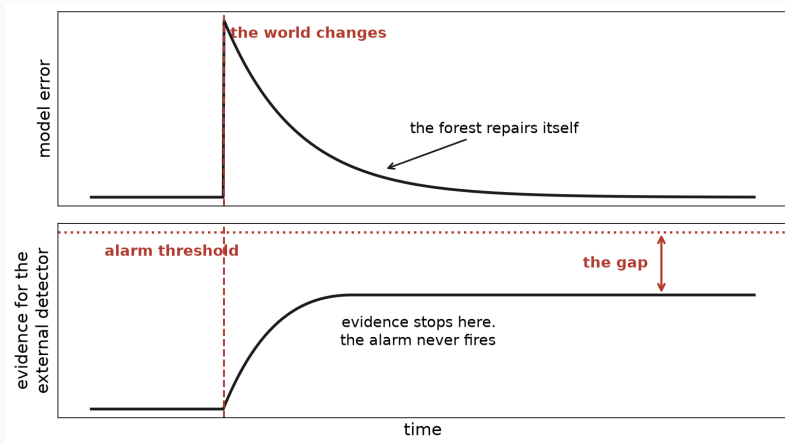
Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

Two machines, and only one of them talks to you

- A **model** predicts, learns as it goes, and **repairs itself** when the world stops matching what it learned.
- A separate **watchdog** reads the model's mistakes and raises an alarm once they pile up.
- The alarm is the only thing a human ever sees. It is what triggers the audit, the retraining, the pause.

Nobody watches the model. The watchdog does.

The repair destroys the evidence

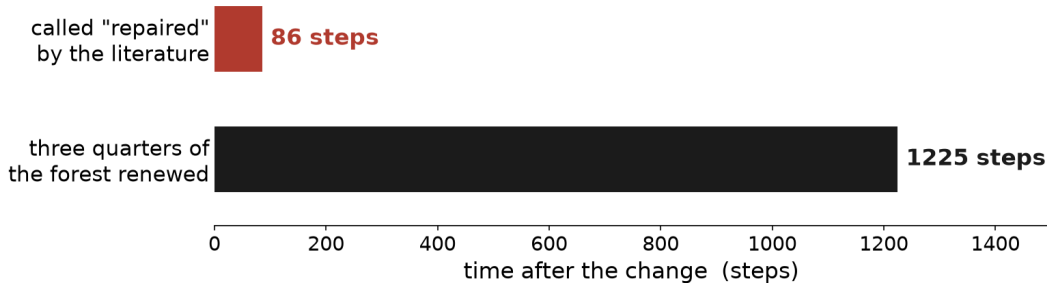


3 alarms out of 2 000 runs

re-running the source paper's own benchmark, here, on our machines

- One research field measures how fast a model recovers. Another measures how fast an alarm fires.
- Neither has measured them against each other.

What we found: the clock was wrong



median over 100 runs at $\Delta e = 0.24$, $M = 10$

The accepted date marks the first tree out of ten, not a repaired forest.

Where we stand

- The paradox is **real and reproducible**, and we can now say exactly when the evidence falls short.
- The measure the field relies on is **biased, always the same way**. Any conclusion drawn from it inherits that bias.
- **Next:** watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.